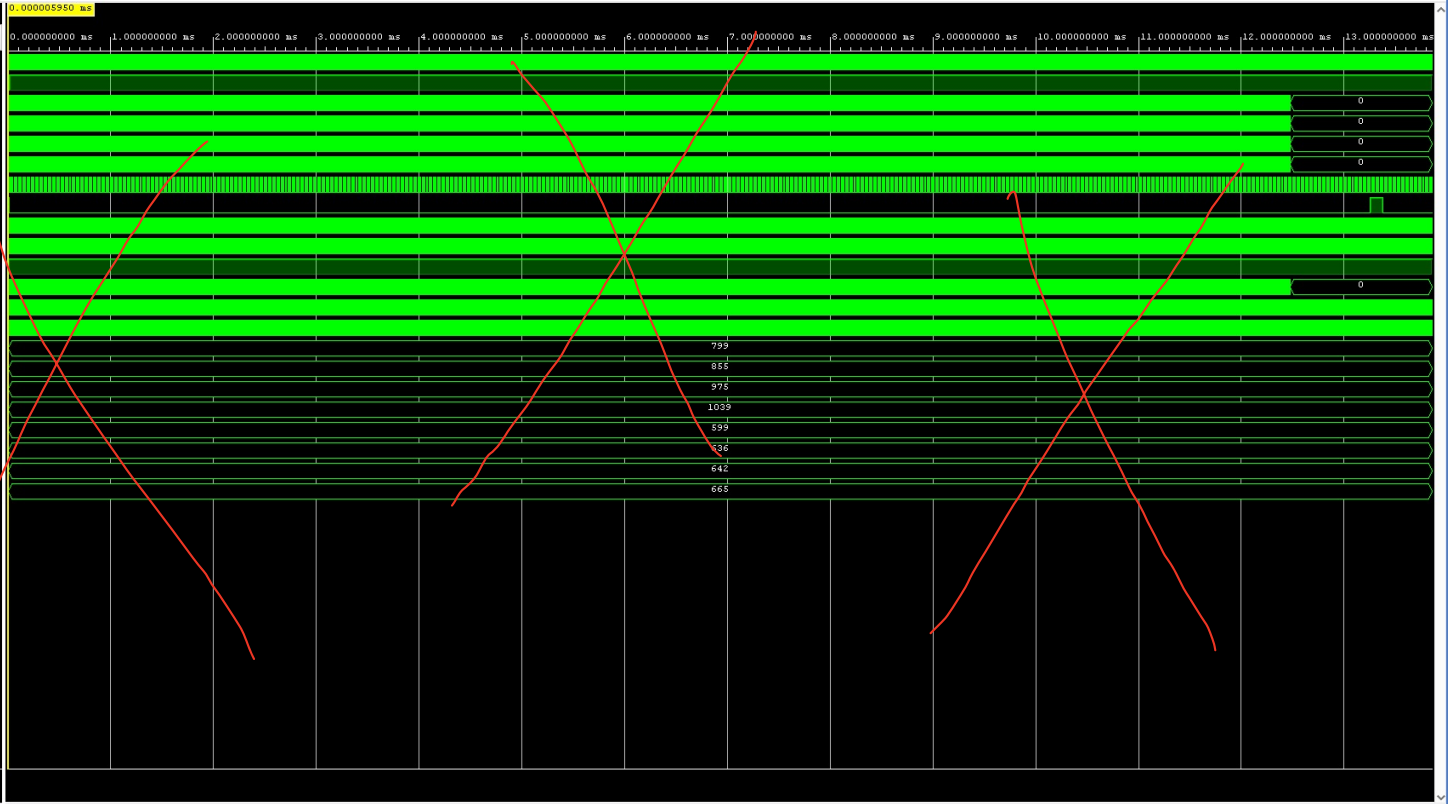
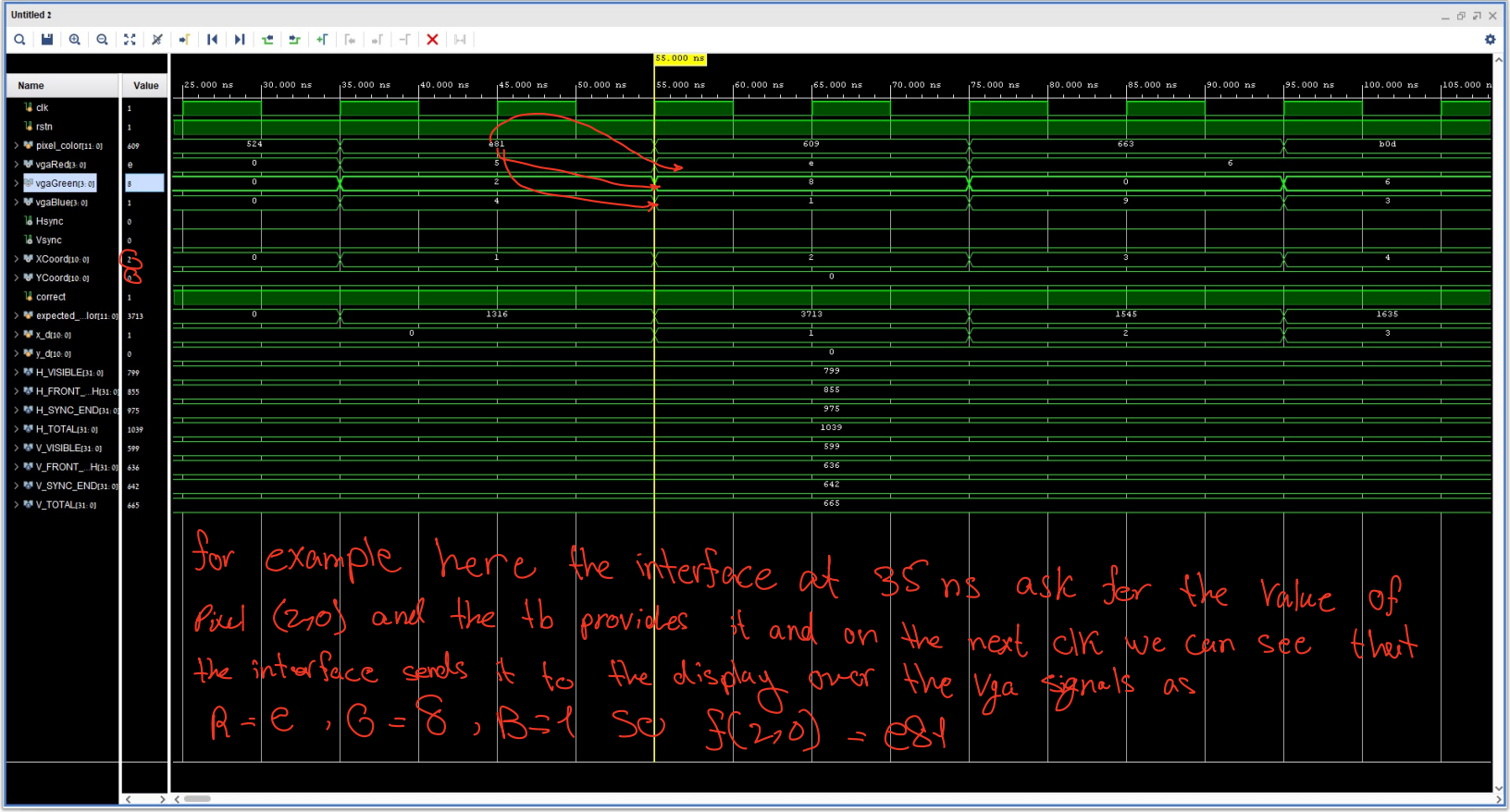
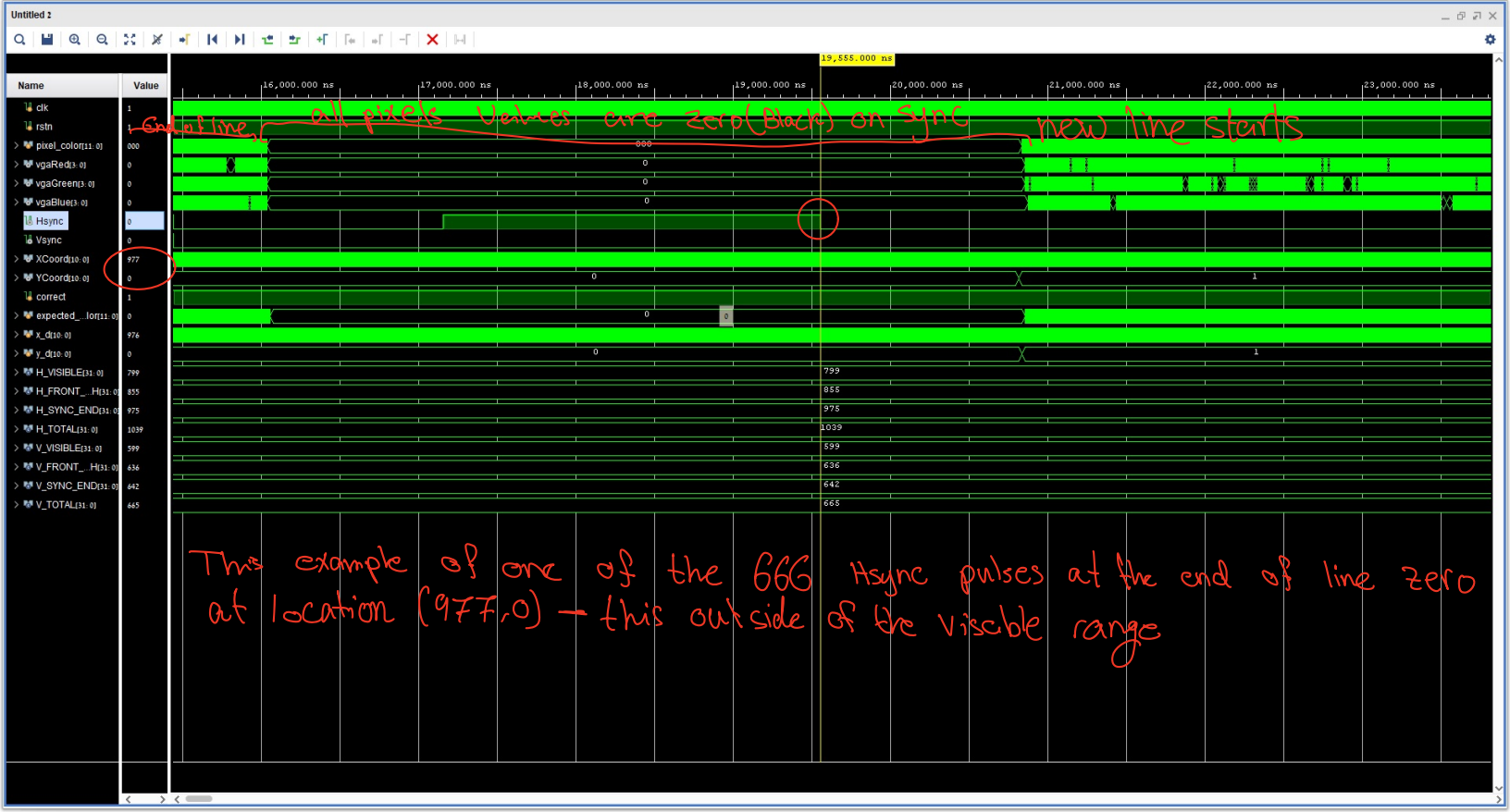
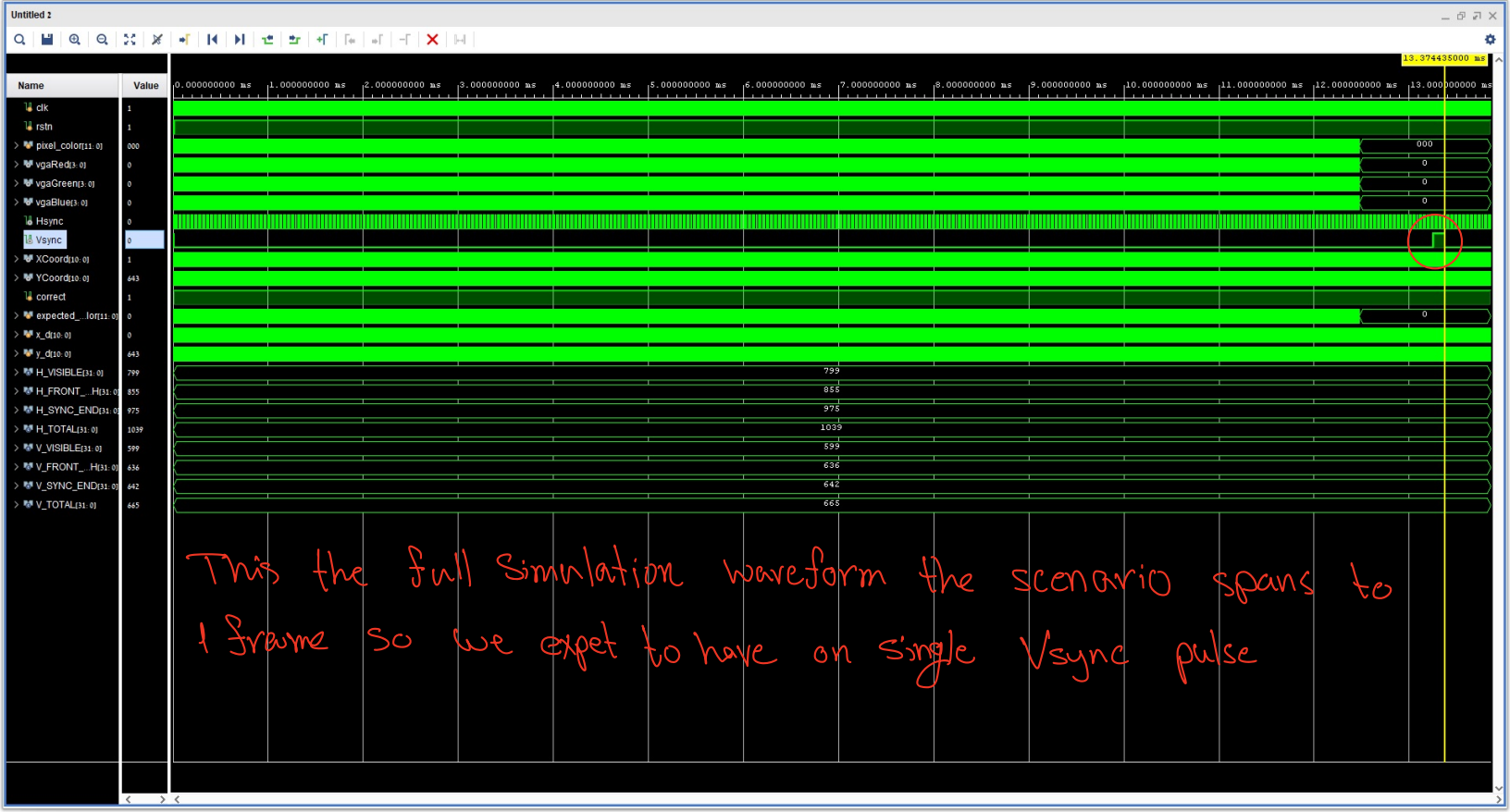
Untitled 2

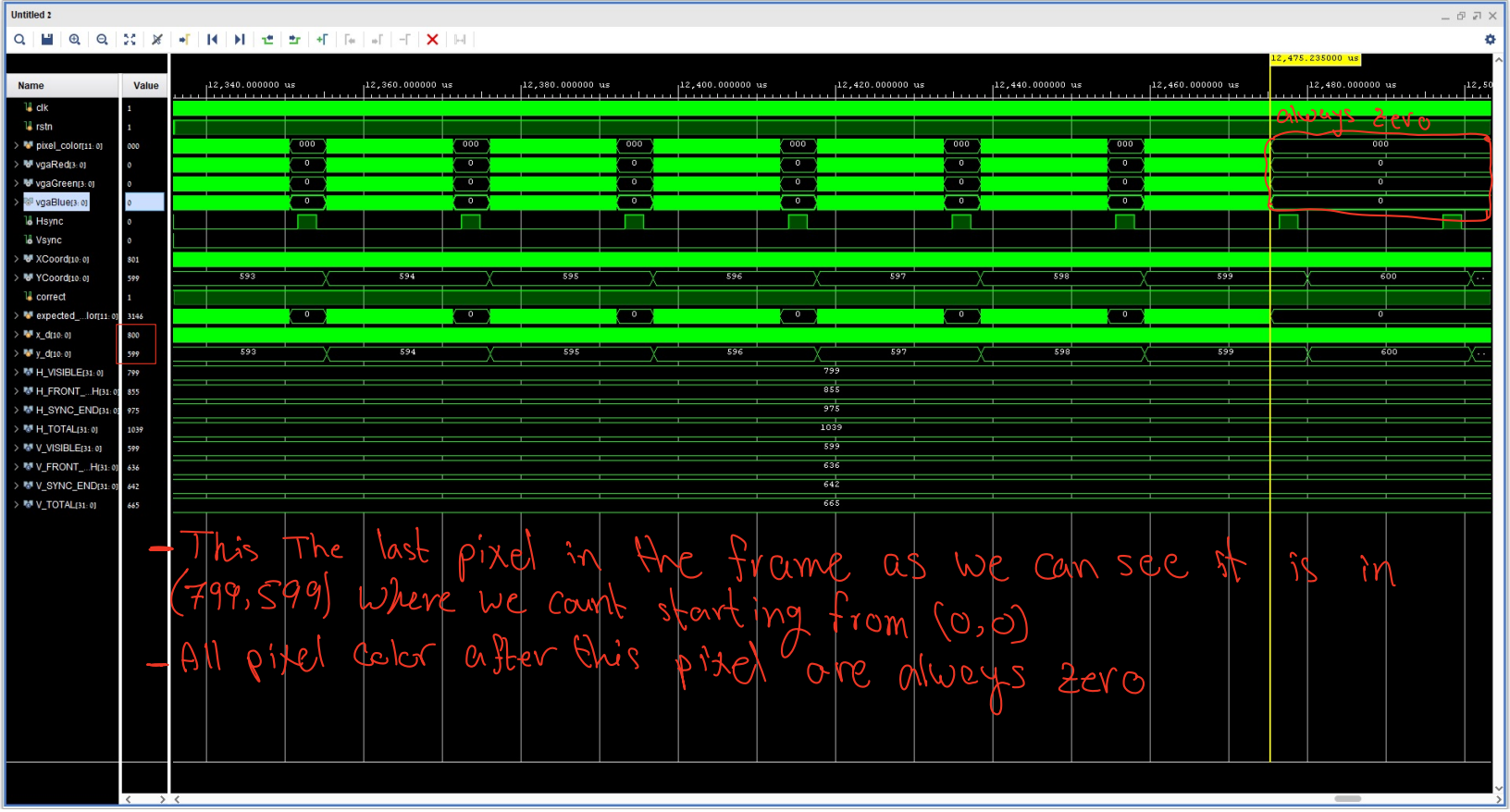
Name	Value
clk	1
rstn	0
pixel_color[11:0]	0
vgaRed[3:0]	0
vgaGreen[3:0]	0
vgaBlue[3:0]	0
Hsync	0
Vsync	0
XCoord[10:0]	0
YCoord[10:0]	0
correct	1
expected_...lor[11:0]	0
x_d[10:0]	0
y_d[10:0]	0
H_VISIBLE[31:0]	799
H_FRONT_...H[31:0]	855
H_SYNC_END[31:0]	975
H_TOTAL[31:0]	1039
V_VISIBLE[31:0]	599
V_FRONT_...H[31:0]	636
V_SYNC_END[31:0]	642
V_TOTAL[31:0]	665




Untitled 1
55.000 ns
Name
Value
clk
rstn
pixel_color[11:0]
vgaRed[3:0]
vgaGreen[3:0]
vgaBlue[3:0]
Hsync
Vsync
XCoord[10:0]
YCoord[10:0]
correct
expected_...lor[11:0]
x_d[10:0]
y_d[10:0]
H_VISIBLE[31:0]
H_FRONT_...H[31:0]
H_SYNC_END[31:0]
H_TOTAL[31:0]
V_VISIBLE[31:0]
V_FRONT_...H[31:0]
V_SYNC_END[31:0]
V_TOTAL[31:0]
for example here the interface at 35 ns ask for the value of
Pixel (2,0) and the tb provides it and on the next clk we can see that
the interface sends it to the display over the Vga signals as
R = e , G = 8 , B = 1 so f(2,0) = e81


Untitled 2
Name
Value
clk
rstn
pixel_color[11:0]
vgaRed[3:0]
vgaGreen[3:0]
vgaBlue[3:0]
Hsync
Vsync
XCoord[10:0]
YCoord[10:0]
correct
expected_...lor[11:0]
x_d[10:0]
y_d[10:0]
H_VISIBLE[31:0]
H_FRONT_...H[31:0]
H_SYNC_END[31:0]
H_TOTAL[31:0]
V_VISIBLE[31:0]
V_FRONT_...H[31:0]
V_SYNC_END[31:0]
V_TOTAL[31:0]
19,555.000 ns
16,000.000 ns
17,000.000 ns
18,000.000 ns
19,000.000 ns
20,000.000 ns
21,000.000 ns
22,000.000 ns
23,000.000 ns
end of line
all pixels values are zero(Black) on sync
new line starts
This example of one of the 666 Hsync pulses at the end of line zero at location (977,0) - this outside of the visible range


Untitled 2
Name
Value
clk
rstn
pixel_color[11:0]
vgaRed[3:0]
vgaGreen[3:0]
vgaBlue[3:0]
Hsync
Vsync
XCoord[10:0]
YCoord[10:0]
correct
expected_...lor[11:0]
x_d[10:0]
y_d[10:0]
H_VISIBLE[31:0]
H_FRONT_...H[31:0]
H_SYNC_END[31:0]
H_TOTAL[31:0]
V_VISIBLE[31:0]
V_FRONT_...H[31:0]
V_SYNC_END[31:0]
V_TOTAL[31:0]
13.374435000 ms
This the full simulation waveform the scenario spans to 1 frame so we expet to have on single Vsync pulse


Untitled 2
Name
Value
clk 1
rstn 1
pixel_color[11:0] 000
vgaRed[3:0] 0
vgaGreen[3:0] 0
vgaBlue[3:0] 0
Hsync 0
Vsync 0
XCoord[10:0] 801
YCoord[10:0] 599
correct 1
expected_...lor[11:0] 3146
x_d[10:0] 800
y_d[10:0] 599
H_VISIBLE[31:0] 799
H_FRONT_...H[31:0] 855
H_SYNC_END[31:0] 975
H_TOTAL[31:0] 1039
V_VISIBLE[31:0] 599
V_FRONT_...H[31:0] 636
V_SYNC_END[31:0] 642
V_TOTAL[31:0] 665
12,475.235000 us
always zero
- This The last pixel in the frame as we can see it is in (799,599) where we count starting from (0,0)
- All pixel color after this pixel are always zero